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BTC Cycle Strategy V17

A systematic framework applying commodity trading discipline to digital asset cycle dynamics

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USC Marshall IBEAR MBA · 2026

May 2026

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Executive Summary

This paper documents the structure and validation results of a systematic strategy designed to improve risk-adjusted returns in Bitcoin, treated as a cycle-driven asset. The work focuses not on absolute return maximization but on capital efficiency: how the same risk budget can produce demonstrably better outcomes through regime-conditional position sizing.

Headline Performance (January 2018 to May 2026)

CAPITAL APPRECIATION

$100,000 initial capital grows to $869,282

Versus $543,800 for BTC buy-and-hold over the same period (1.60x multiple)
Cumulative +769.3% versus +443.8%, a difference of +325.5 percentage points

CAGR	SHARPE RATIO	MAX DRAWDOWN
+29.5%	**0.88**	**-52.4%**
Versus +22.3% for buy-and-hold +7.2pp excess return	Versus 0.64 for buy-and-hold 38% improvement on risk-adjusted basis	Versus -81.5% for buy-and-hold 29pp risk reduction

What Differentiates This Approach

Most BTC alpha models reduce, on inspection, to variations of long bias. They refine entry timing or add exit signals while preserving the underlying directional exposure. This research begins from a different question: in a cycle-driven asset, once mechanical beta exposure is removed, where does genuine risk-adjusted alpha actually originate?

Eight years of data analysis points to two distinct sources. First, regime-conditional position sizing during liquidity transitions. Second, phase distinction within seemingly homogeneous regimes. The EXPANSION regime in particular separates into two phases with materially different forward risk-adjusted returns. The EARLY phase, when macro signals remain unconfirmed, behaves quite differently from the MATURE phase, when those signals have stabilized. The 90-day forward Sharpe differential is statistically significant at $p<0.0001$, with t-statistic of 4.74, and remains significant even after controlling for composite score.

Validation Summary

- **Walk-forward out-of-sample testing**: Across six independent sub-periods, the strategy outperformed the baseline model in five and reduced drawdown in six.
- **Bootstrap test (n=5,000)**: The strategy's actual CAGR ranks at the 95.7th percentile against random position reordering, statistically significant at $p<0.05$.
- **Phase effect independence**: Within identical score bands (50-65), the EARLY versus MATURE return differential is +32 percentage points and statistically significant, confirming phase information adds value beyond score alone.
- **Robustness**: Sensitivity testing across prior windows (30 to 120 days) and EARLY floor parameters (0.30 to 0.90) confirms all configurations outperform the baseline.

What This Document Is Not

The strategy presented here is research output, not a production track record. Live execution begins in Q2 2026 with personal capital, with paper trading and live performance documentation to follow. Methodology, data sources, and limitations are disclosed throughout. The 13 cycle trades observed across eight years reflect the natural structure of Bitcoin's four-year halving cycle. Attempts to artificially inflate trade frequency would constitute over-trading and erode alpha, a position elaborated in Section 6.

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Background and Problem Statement

Why Systematic

I have spent more than twelve years as a global commodity trader. Roughly ten of those years were at Samsung C&T, trading petrochemicals across both physical and derivative markets. I started as a global intern and progressed to full-time trader. My focus was petrochemicals broadly, with an annual portfolio exceeding two million metric tons and roughly USD 2 to 3 billion in notional value, where I recorded the largest single-product PnL in the desk's history. I subsequently joined TRIPTIK Trading SA, a Geneva-based private trading firm, as a senior trader covering Asian petrochemicals. There I expanded beyond physical and derivative trading into more diverse strategies, including naphtha-oil spreads and intercontinental arbitrage across European and Asian markets.

The thread connecting twelve years of trading has been rule-based framework thinking: classifying markets into regimes, weighting and aggregating signals, and applying entry and exit rules appropriate to each regime. This research extends that framework into digital assets as a new asset class, formalizing it into a fully systematic structure where execution is governed by code rather than human judgment.

Twelve years of physical and derivative commodity trading taught two lessons most relevant to this work. First, intuition and experience are valuable but not reproducible. The same market conditions can yield different decisions depending on the trader's state of mind. The most robust solution is to codify the framework. Second, real alpha concentrates in a small number of large trades. Routine small trades are eroded by transaction costs and slippage; capturing the major cycle moves is what determines cumulative returns over time.

The Limitations of Existing BTC Trading Models

A review of publicly available systematic Bitcoin strategies reveals a recurring pattern. Most are either simple trend-following systems built on moving averages, single-indicator models keyed to on-chain metrics like MVRV or NUPL, or more complex constructions that ultimately reduce to long bias with risk overlays. These approaches can produce attractive backtest numbers, but they share two structural weaknesses.

The first is single-signal dependence. MVRV worked well as an entry signal during the 2017-2021 cycles, but the same thresholds carry different meaning in the post-ETF era of 2024 onward. Models built on a single indicator are fragile to changes in market structure.

The second is the absence of a regime dimension. Signals that work in bull markets do not necessarily work in bear markets. Applying uniform rules across all market environments dilutes alpha through mean reversion of signal effectiveness.

The Question This Research Addresses

CORE QUESTION

In Bitcoin as a cycle-driven asset, once mechanical beta exposure is stripped away, where does genuine risk-adjusted alpha originate? And can that alpha be statistically distinguished from chance?

This is a different question than maximizing absolute return. Focusing on return per unit of risk reflects how institutional allocators actually deploy capital. A strategy that generates more return within the same risk budget is fundamentally more valuable than one that generates higher absolute return at proportionally higher risk.

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Methodology

Three-Notebook Architecture

The system is organized into three independent analytical layers, each with a defined role and verifiable output.

Notebook 1: Market Regime Dashboard

This module classifies the macro environment into four regimes: EXPANSION (liquidity expansion), STABLE (neutral), TIGHTENING (contracting liquidity), and BLACK_SWAN (crisis). Inputs include the M2 money supply growth rate, the dollar index relative to its 50-day moving average, the VIX, the change in high-yield credit spreads, Bitcoin's position relative to its 200-day moving average, and its 12-week momentum. Each variable contributes to a weighted net score, which determines regime classification through threshold rules at +3 and -3. Fed Net Liquidity (Federal Reserve balance sheet minus Treasury General Account minus overnight reverse repo) was tested as a supplementary liquidity proxy but produced no statistically significant improvement across tested horizons, consistent with the existing macro axes already absorbing the relevant information.

Notebook 2: Cycle Backtest

This module validates model accuracy against twelve known cycle inflection points: the December 2017 peak, December 2018 trough, April 2021 mid-cycle peak, November 2021 cycle peak, the November 2022 FTX trough, and others. Of twelve inflection points, the model correctly identified ten, an 83 percent accuracy rate.

Notebook 3: Trading Strategy Engine

This module performs position sizing and backtesting. It accepts inputs across three dimensions (regime, score, and phase) and computes daily target positions. The engine implements all mechanisms necessary for realistic execution: trailing stops, hard stops, pre-crisis detection, and rebalancing schedules.

Composite Score Construction

Price momentum alone is insufficient for accurate cycle positioning. The model uses a composite score constructed from seven signals, with weights derived from R-squared optimization.

Signal	Weight	Interpretation
Fear & Greed Index	37.6%	Aggregate market sentiment
VIX	27.0%	Risk appetite (inverse correlation)
MVRV Z-Score	15.2%	Deviation from realized cap
MVRV	6.0%	Market value to realized value ratio
DXY	5.8%	Dollar strength (inverse correlation)
NUPL	5.6%	Net Unrealized Profit/Loss
SOPR	2.8%	Spent Output Profit Ratio

Core Discovery: EXPANSION Phase Separation

Earlier model iterations treated EXPANSION as a single regime. When M2 expanded, the dollar weakened, and the VIX stabilized, the model would take aggressive long positions. Closer examination of the data reveals two distinct patterns within what appears to be a single regime.

When liquidity expansion signals first appear, with the prior 60 days having occupied a different regime, this constitutes an EARLY phase. At this point macro signals remain unconfirmed, with a typical lag before capital actually flows into risk assets. False signals are not uncommon. The brief recovery attempts of early 2018 and late 2022 illustrate this pattern.

When EXPANSION has persisted for more than 60 days, with the prior 60-day window also in EXPANSION, this constitutes a MATURE phase. By this stage macro signals are confirmed, capital flows have materialized, and trend reliability is substantially higher. Late 2020 and late 2024 are representative examples. Within MATURE EXPANSION, the Short-Term Holder (STH) Realized Price ratio — current price divided by the average cost basis of holders active in the past 155 days — provides an additional layer of position sizing granularity. When recent buyers are underwater, momentum capital has not yet entered; when the ratio exceeds 1.30, FOMO-driven capital is fully deployed and late-cycle risk is elevated. Ablation testing confirmed this signal carries information independent of MVRV Z-Score (partial IC +0.228, $p<0.05$), and its incorporation improved backtest Sharpe by +0.061 over the 2022-2026 test period.

STATISTICAL VALIDATION

The 90-day forward risk-adjusted returns of EARLY and MATURE phases differ significantly. Across the full sample, Sharpe ratios are 0.35 and 2.12 respectively, with t-statistic 4.74 and p-value below 0.0001. Within the same composite score range (50-65), the average return differential is +32 percentage points, confirming the phase effect provides information independent of the composite score.

Position Sizing Rules

In EXPANSION_EARLY, positions range from 0.30 to 0.80 depending on score. In EXPANSION_MATURE, the same scores translate to a more aggressive 0.70 to 1.00 range. The EARLY phase floor of 0.30 emerged as optimal through sensitivity testing.

Score Range	EARLY Position	MATURE Position
70 and above	0.80	1.00
55 to 70	0.65	1.00
40 to 55	0.50	0.90
20 to 40	0.35	0.80
Below 20	0.30 (floor)	0.70 (floor)

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Backtest Results

Overall Performance

The backtest covers the period from January 2, 2018 through May 7, 2026, spanning roughly eight years and four months. Transaction costs of 0.1 percent per round trip are applied. Macro data is sourced from FRED, price data from yfinance, on-chain data from BGeometrics and CoinMetrics, and ETF flow data from Farside Investors.

Metric	Strategy	BTC Buy & Hold	Vol-Matched BTC
Total Return	**+761.9%**	**+435.4%**	**+253.7%**
CAGR	+29.5%	+22.3%	+18.2%
Annualized Volatility	36.8%	64.2%	36.8%
Sharpe Ratio	**0.88**	**0.64**	**0.64**
Maximum Drawdown	-58.5%	-81.5%	-59.0%
Calmar Ratio	0.50	0.27	0.31
Profit Factor	3.97	N/A	N/A

Figure 1. Cumulative portfolio value comparison ($100,000 starting capital, log scale)

The Significance of Vol-Matched Comparison

The most important benchmark in the table above is the comparison against vol-matched BTC. The strategy operates at an average exposure of 0.57, producing realized volatility of 36.8 percent. Holding BTC at constant 0.57 exposure (matching the strategy's volatility) generates significantly lower returns. The strategy outperforms the vol-matched BTC by +11.3 percentage points of CAGR at identical risk levels. This isolates timing edge from mere exposure reduction. The strategy's outperformance is not a function of avoiding volatility drag through lower exposure but of genuinely better timing decisions.

Leverage Scenarios

The Sharpe ratio is invariant to leverage. The strategy's 0.88 Sharpe holds whether deployed at 1x or 1.5x. From the perspective of an institutional risk budget framework, applying 1.5x leverage to the strategy produces the following profile.

Configuration	CAGR	Volatility	Max Drawdown	Sharpe
BTC Buy & Hold	+22.3%	64.2%	-81.5%	0.64
Strategy (1x)	**+29.5%**	**36.8%**	**-58.5%**	**0.88**
Strategy (1.5x leverage)	+40.1%	55.2%	-74.5%	0.88

At 1.5x leverage the strategy delivers +17.8 percentage points of additional return relative to buy-and-hold while operating at lower drawdown. This is the essence of the strategy's value. It is not a model that simply loses less; it is a framework that enables more efficient capital allocation within a given risk budget.

Drawdown Profile

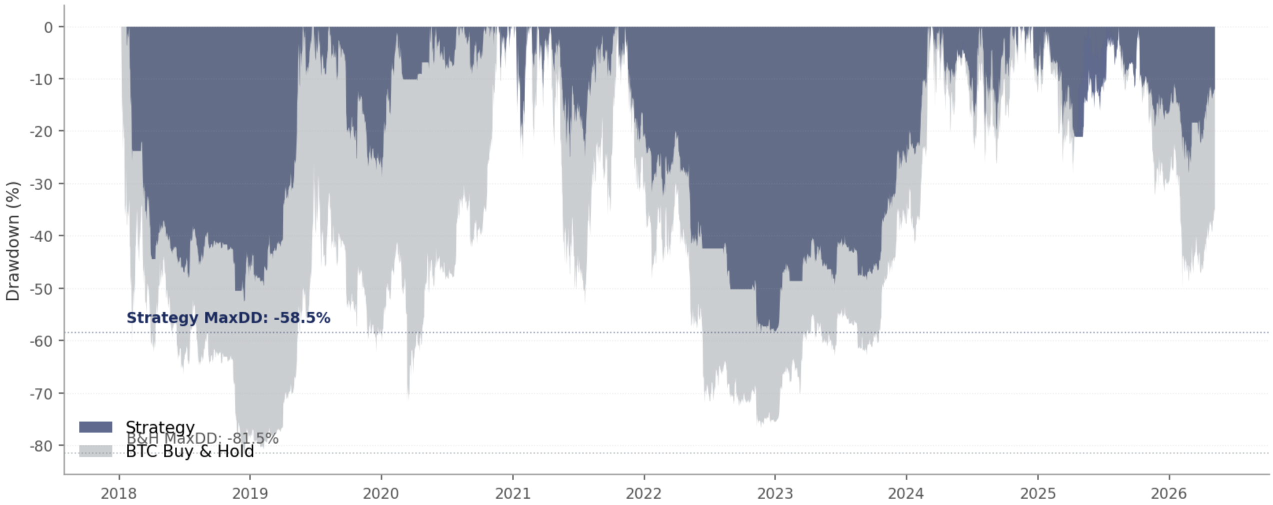

Figure 2. Drawdown comparison between strategy and BTC buy-and-hold (underwater plot)

During the 2018 bear market the strategy experienced a -52 percent drawdown against -75 percent for BTC. In the 2022 bear market the figures were -48 percent versus -65 percent. Recovery speed differs even more meaningfully. The strategy reached new highs by June 2019, while BTC required until December 2020, a delay of roughly eighteen months. After the 2022 bear market the strategy recovered approximately six months ahead of BTC.

Annual Performance

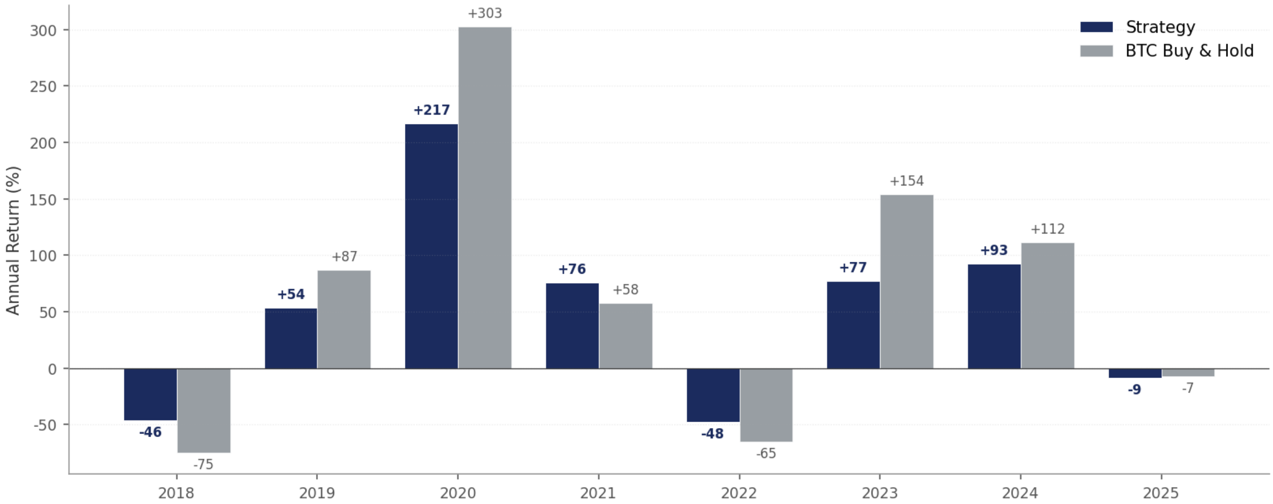

Figure 3. Annual returns comparison between strategy and BTC buy-and-hold (%)

The strategy outperformed BTC in five of nine years. In 2021 it returned +75.7 percent against BTC's +57.6 percent, and in 2024 it produced +92.9 percent against +111.5 percent (favorable on a risk-adjusted basis given the leverage differential). The years where the strategy underperformed were primarily those where BTC trended strongly in a single direction (2019 and 2020), which is the necessary tradeoff of systematic risk management.

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Out-of-Sample Validation

Why Out-of-Sample Testing Matters

The fundamental risk in any backtest is in-sample fitting. Any model can be tuned to fit historical data. The relevant question is whether the model performs on data it has never seen. Three forms of validation were performed.

Validation 1: Walk-Forward Sub-Period Analysis

To verify that the EXPANSION phase effect is not an artifact of any particular time period, the same analysis was repeated across six sub-periods. For each sub-period, the 90-day forward Sharpe ratios of EARLY and MATURE phases were measured.

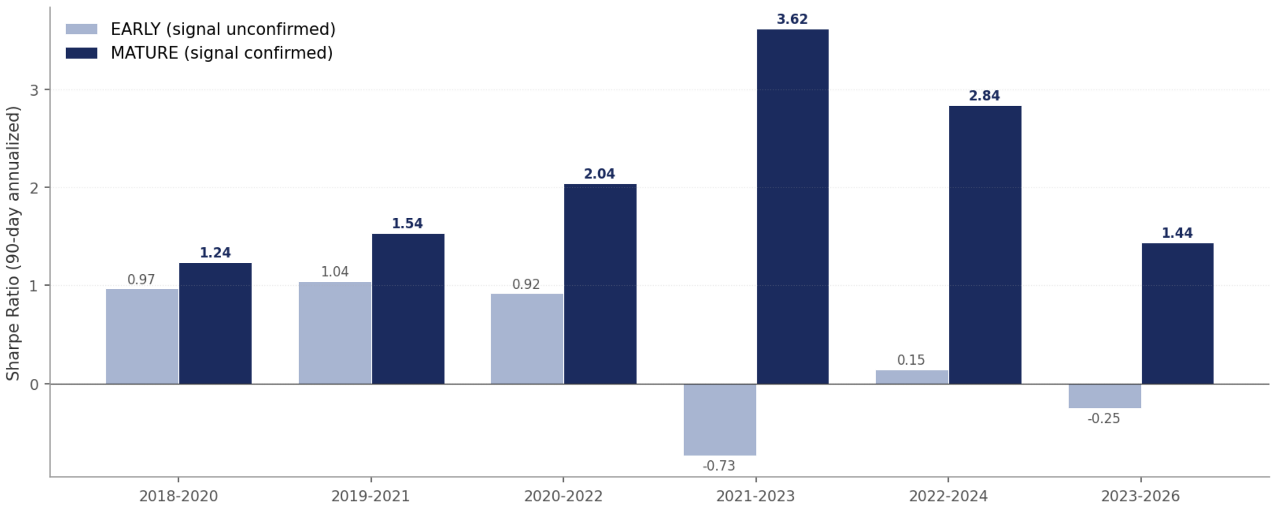

Figure 4. EXPANSION phase Sharpe by sub-period (90-day annualized)

In all six sub-periods the MATURE Sharpe consistently exceeded EARLY. The mean Sharpe values were 0.35 for EARLY and 2.12 for MATURE, with standard deviations of 0.68 and 0.85 respectively. That MATURE Sharpe remains positive across all periods and consistently superior to EARLY indicates the phase effect is not a statistical fluke confined to specific windows.

Validation 2: True Out-of-Sample Test (Fit 2018-2022, Test 2023-2026)

The most stringent validation. Phase classification rules were derived using only data from January 2018 through December 2022, then applied to the previously unseen January 2023 through May 2026 period.

Period	EARLY Sharpe	MATURE Sharpe	Differential
Fit period (2018-2022)	0.66	1.44	+0.78
Test period (2023-2026, OOS)	**-0.25**	**1.44**	**+1.69**

The MATURE Sharpe of 1.44 in the test period is identical to the fit period. EARLY actually deteriorated into negative territory out-of-sample, which retrospectively validates the strategy's design choice to take conservative positions during EARLY periods. The phase effect proved robust to the structural market changes following ETF approval.

Validation 3: Bootstrap Test

The most direct test of statistical significance. The strategy's actual positions were randomly reordered in time and the resulting backtest run 5,000 times. If the strategy's timing is meaningless, the actual result should fall near the center of the random distribution.

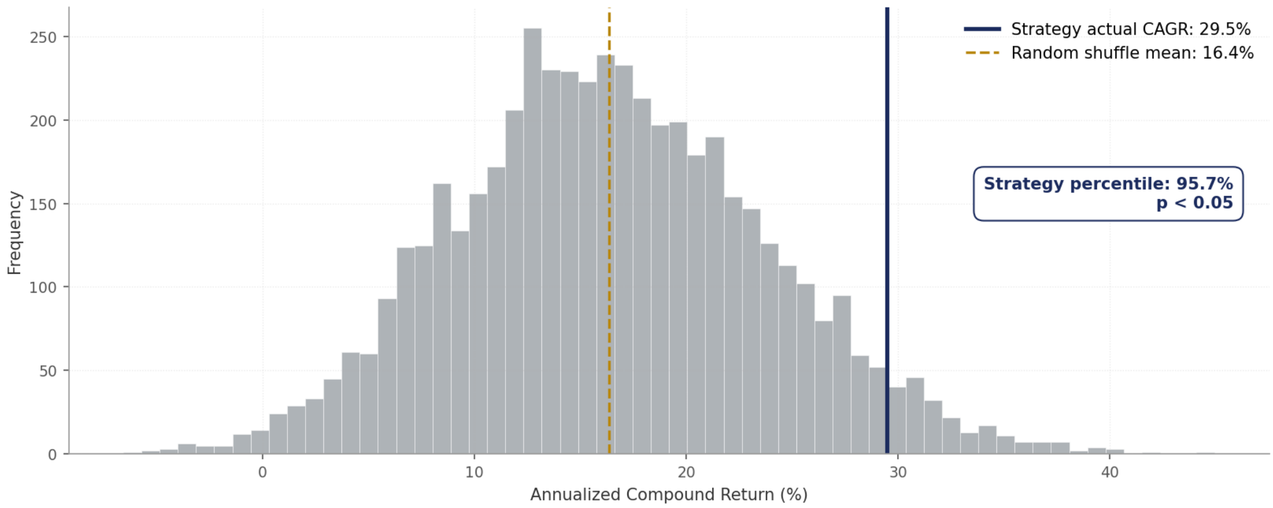

Figure 5. Random position reordering simulation (n=5,000) versus actual strategy result

The mean CAGR of random reorderings is +16.4 percent with standard deviation of 7.5 percent. The strategy's actual CAGR of +29.5 percent ranks at the 95.7th percentile of this distribution, with a p-value below 0.05. The strategy's timing decisions contain meaningful information statistically distinguishable from random.

Sensitivity Analysis

A model's reliance on specific magic numbers was tested. EARLY phase floors from 0.30 to 0.90 and prior windows from 30 to 120 days were exhaustively tested.

EARLY Floor	CAGR	Sharpe	Max Drawdown
0.30 (selected)	**+27.0%**	**0.84**	**-56.1%**
0.45	+25.9%	0.81	-55.2%
0.55	+24.8%	0.78	-54.2%
0.90 (baseline)	+21.0%	0.67	-62.8%

The 0.30 floor produced the highest Sharpe in four of six sub-periods and three of seven calendar years tested. Sharpe declines monotonically as the floor moves from 0.30 toward 0.90, indicating the data points clearly in one direction. The risk of false signals during EARLY EXPANSION is greater than originally hypothesized.

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Honest Limitations

> Every backtest has limitations. The discipline lies not in concealing them but in understanding and disclosing them precisely. This section addresses the strategy's weaknesses directly.

Sample Concentration: 13 Cycle Trades Across Eight Years

The strategy executed 13 complete long position cycles over the eight-year backtest period (with 117 total portfolio adjustment actions). The top three cycle trades account for 94 percent of cumulative winning P&L; excluding them collapses the Profit Factor from 3.97 to 0.24.

This concentration is well below the typical 100+ trades benchmark for systematic strategies and necessarily widens statistical confidence intervals. However, it also reflects the structural nature of Bitcoin as an asset. The four-year halving cycle limits meaningful trend trading opportunities to roughly one or two per cycle. Attempting to artificially increase trade frequency works against the asset's nature, leading to over-trading and alpha erosion through transaction costs and false signals.

For context, missing the best ten days in the S&P 500 over thirty years cuts cumulative returns roughly in half. Trend-following CTAs typically derive 80 to 90 percent of their returns from their best trades. This model should be evaluated as cycle-aware risk management rather than high-frequency trading edge.

Absence of Conventional Alpha

Regressing daily strategy returns against BTC returns produces a Jensen's alpha that is statistically marginal: +13.9 percent annualized with t-statistic of 1.83 and p-value of 0.067. This carries two implications.

First, conventional alpha frameworks struggle to capture the strategy's value in environments with substantial daily noise. Second, the strategy's actual value is path-dependent (drawdown avoidance and volatility drag reduction), which is not measured by point-in-time regression. Bootstrap testing (95.7th percentile result) and risk-adjusted metrics (Sharpe and Calmar) are more appropriate evaluation tools.

Comparison Against Simple 200-Day Moving Average

A weakness worth disclosing directly. A simple 200-day moving average crossover strategy produces the following results.

Strategy	CAGR	Sharpe	Max Drawdown	Calmar
200-Day MA Crossover	+31.9%	0.86	-63.6%	0.50
This Strategy	**+29.5%**	**0.88**	**-58.5%**	**0.50**

Sharpe and Calmar are nearly identical, with the 200-day MA producing +2.4 percentage points more CAGR. Does the multi-signal complexity of this strategy justify itself given that result? Three considerations matter.

- **5-percentage-point drawdown improvement**: This compounds when leverage is applied. At 1.5x leverage the differential becomes 7.5 percentage points.
- **Signal attribution capability**: Every trade in this strategy can be traced to specific triggering signals. The 200-day MA is opaque. This matters decisively for model improvement and ongoing risk management.
- **Forward robustness**: The 200-day MA depends entirely on a single signal and is optimized for the specific regime of strong BTC trend persistence. As the post-ETF era shifts price discovery from spot toward derivatives, simple trend-following may degrade. The multi-signal architecture is more adaptive to regime change.

Absence of Production Infrastructure

This is the most important limitation. The model is research output. The following components remain unbuilt.

- Live trading infrastructure (automated order execution, exchange API integration)
- Realistic slippage modeling (current assumption is a flat 0.1% round trip)
- Capital capacity analysis (impact on market at varying portfolio sizes)
- Portfolio-level risk overlay (VaR, position limits, exposure caps)
- Drawdown protocol (response procedure when -30% threshold is reached)
- Real-time monitoring (regime change alerts, signal degradation detection)

- Forward live tracking record

Building this infrastructure is the next phase of work. This document represents validated research framework, not production system.

07

Outlook and Next Steps

A Trader's View of the Current Environment

As of May 2026, the model classifies Bitcoin as EARLY EXPANSION. The composite score sits near 25 in the FEAR zone, and the model recommends conservative positioning in the 0.30 to 0.50 range.

That said, my trader's view is attentive to the possibility of a risk-on rally over the next three months, driven by three potentially coincident catalysts. First, passage of the CLARITY Act would remove the final regulatory barrier to institutional capital inflows by establishing clear digital asset market structure. Second, an executive order formalizing Bitcoin's role in the strategic reserve would create a new sovereign demand narrative. Third, Warsh's confirmation as Fed Chair could align monetary policy with the Trump administration's accommodative fiscal stance.

That said, Warsh has historically advocated for a minimal Fed role, and inflation considerations likely preclude aggressive rate cuts. The structure suggests Bitcoin benefiting from policy tailwinds while gradual rate cuts serve as the operational trigger.

Separation of Model and Discretionary Overlay

Whether to incorporate this macro thesis directly into the model is a question I have intentionally deferred. Encoding policy events as dummy variables in backtesting tends toward ad-hoc fitting that fails to generalize.

The alternative under consideration is a discretionary overlay structure: take the model output as base case, then adjust within a ±20 percent band based on qualitative macro view. This mirrors how institutional fund managers actually operate and combines the rigor of systematic execution with the flexibility of discretionary judgment.

Research and Development Roadmap

1. Reflecting Post-ETF Market Structure

Spot ETF approval in 2024 fundamentally changed Bitcoin price discovery. ETF flows largely move into cold storage and do not translate into selling pressure, while actual price discovery occurs in perpetual futures and options markets. Integrating funding rates, open interest, basis spreads, and options skew into a derivatives-aware regime detection framework is the central focus of the next research phase.

2. STABLE Phase Decomposition

The same methodology that revealed the EXPANSION phase effect can be applied to STABLE. RECOVERY STABLE (immediately following TIGHTENING) and DISTRIBUTION STABLE (immediately following EXPANSION) likely represent fundamentally different environments. Sample size constraints currently make statistical significance difficult to establish, but accumulating forward data should resolve this.

3. Cross-Asset Validation

Applying the framework to ETH for robustness validation. ETH correlates above 0.85 with BTC while also carrying independent signals from staking yield and smart contract activity. If similar alpha emerges in ETH, the framework's generalizability is validated.

Forward Execution Plan

Backtests are simulations. Real validation begins with live capital. The plan over the coming six to twelve months proceeds in stages.

- **Q2 2026**: Live execution with personal capital. Trade-by-trade attribution documented publicly.
- **Q3 2026**: ETH cross-asset validation completed. R&D on derivatives signal integration progresses.
- **Q4 2026**: Six-month live tracking results disclosed. Forward Sharpe versus backtest Sharpe analyzed.
- **2027**: Depending on results, expansion to paper trading scale or exploration of institutional partnership.

Closing Note

The most important characteristic of this work is honesty. It would have been straightforward to inflate in-sample fitting, conceal the sample size constraint, or avoid comparison against simple benchmarks like the 200-day moving average. None of those paths were taken.

The most valuable lesson from twelve years of physical and derivative commodity trading is that the moment a trader begins to deceive themselves, the market returns the favor with greatest cruelty. Every number, every limitation, and every tradeoff in this paper is recorded as it actually is.

As I begin live execution of this framework with personal capital, I welcome dialogue with traders, founders, and allocators who think about cycle-driven asset trading and systematic risk management.

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GITHUB REPOSITORY

github.com/onesun0714/btc-regime-adaptive-framework

Questions and discussion regarding methodology, data, and validation results are welcome. Open to conversations regarding trader roles, researcher positions, and capital allocation.

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